

Wafer-Rack Equal- Q Navier–Stokes Study

Audit, Correction, and Independent Numerical Verification

CFD code review & reproduction

October 1, 2026

Abstract

A two-dimensional incompressible Navier–Stokes solver for flow through a rack of six wafers is reviewed, corrected, and independently re-run. The solver uses a geometry-aligned Marker-and-Cell (MAC) staggered grid with a fractional-step (projection) method, first-order donor-cell upwind advection and explicit second-order diffusion. An equal-rack-flow constraint fixes the inlet velocity of each pitch by a bracketed scalar root solve. Independent reruns reproduce the audited production inlet speeds and hydraulic resistances, confirm discrete mass conservation at roundoff ($\sim 10^{-13}\%$), and recover the resistance ordering $R_{h,\text{Small}} > R_{h,\text{Medium}} > R_{h,\text{Large}}$.

Nomenclature

Symbol	Description	Units (2-D)
<i>Geometry</i>		
W	Tank width	m
H	Tank height	m
N_w	Number of wafers (= 6)	–
P	Wafer pitch (centre-to-centre spacing)	m
t_h	Wafer thickness (= 0.015 00 m)	m
y_0, y_1	Lower / upper wafer extent (0.12, 0.25)	m
w_{in}	Inlet slot width (= 0.015 00 m)	m
y_{cut}	Horizontal flux-evaluation line (= 0.20)	m
<i>Field variables</i>		
u, v	Horizontal / vertical velocity components	m s^{-1}
$\mathbf{u}$	Velocity vector (u, v)	m s^{-1}
$ \mathbf{u} $	Flow speed $\sqrt{u^2 + v^2}$	m s^{-1}
p	Kinematic pressure (p/ρ)	$\text{m}^2 \text{s}^{-2}$
U_{in}	Prescribed inlet (slot) vertical speed	m s^{-1}
<i>Fluid / control parameters</i>		
ν	Kinematic viscosity (= $7.500\,00 \times 10^{-4} \text{ m}^2 \text{s}^{-1}$)	$\text{m}^2 \text{s}^{-1}$
Q_{rack}	Flow rate through the five interior gaps at y_{cut}	$\text{m}^2 \text{s}^{-1}$
Q_0	Equal-flow target (= 0.0055463)	$\text{m}^2 \text{s}^{-1}$
Q_{in}	Total inlet flux at the bottom boundary	$\text{m}^2 \text{s}^{-1}$

Symbol	Description	Units (2-D)
$Q(y)$	Net flux across horizontal face y	$\text{m}^2 \text{s}^{-1}$
Δp	Rack pressure drop ($p_{\text{below}} - p_{\text{above}}$)	$\text{m}^2 \text{s}^{-2}$
R_h	Hydraulic resistance $\Delta p / Q_{\text{rack}}$	s m^{-2}
<i>Discretization</i>		
N_x, N_y	Number of pressure cells in x, y	—
n_{gap}	Fluid cells across each inter-wafer gap (= 20)	—
$\Delta x, \Delta y$	Local cell sizes (nonuniform)	m
Δt	Explicit time step	s
res	Steady-state residual (rel. velocity change)	—
<i>Operators / acronyms</i>		
$\nabla \cdot, \nabla$	Discrete divergence / gradient operators	m^{-1}
MAC	Marker-and-Cell staggered grid	—
FD	(legacy) Finite-difference model	—
CFL	Courant–Friedrichs–Lewy number	—

1 Governing equations and method

The flow solves the 2-D incompressible Navier–Stokes equations

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \nu \nabla^2 \mathbf{u}, \quad \nabla \cdot \mathbf{u} = 0, \quad (1)$$

on a geometry-aligned MAC staggered grid. A fractional-step projection advances the velocity: an explicit predictor (first-order donor-cell upwind advection, second-order nonuniform diffusion) is followed by a pressure Poisson solve and a divergence-free correction,

$$\nabla^2 p = \frac{1}{\Delta t} \nabla \cdot \mathbf{u}^*, \quad \mathbf{u}^{n+1} = \mathbf{u}^* - \Delta t \nabla p. \quad (2)$$

The discrete gradient (correction) and divergence operators are mutually adjoint, so the assembled pressure matrix is exactly $-\nabla \cdot \nabla$ of the operators actually applied; this guarantees discrete mass conservation. The pressure Poisson system is factorized once per grid and reused across all root-finding evaluations. Boundary conditions: no-slip on the tank side/bottom walls and all wafer surfaces, prescribed vertical inlet slots at the bottom, and a $p = 0$ open outlet at the top.

Equal- Q control. For each pitch the inlet speed U_{in} is chosen so the rack flow equals a common target Q_0 . This is a bracketed secant iteration with a bisection safeguard, where every function evaluation is a fully converged CFD solve. It enforces an integral control constraint; it is not a fit to data.

2 Code audit and corrections

The audited bundle was reviewed statically and then exercised. The following non-numerical corrections were applied (none alters the discretization, so the reproduced physics is unchanged):

1. **Hard-coded /mnt/data paths removed.** The driver now imports the solver from its own directory and writes to `$WAFER_OUT` (default: current directory), created lazily so the module imports without filesystem side effects.
2. **Self-test fixed.** `mac_cfd_adaptive.py` previously called `Solver(0.06, 0.0025)`, passing 0.0025 where the integer gap-cell count n_{gap} is expected; this collapsed every gap to a single cell. Corrected to `Solver(0.06, 20)`.
3. **Driver de-duplication.** `run_production_equalq.py` (a byte-identical copy of the audited driver) is now a thin re-export, and the single-case runners import one shared `metrics` implementation.
4. **Stale documentation relabelled.** The legacy FD inlet speeds (0.5605/0.3000/0.1833) are now explicitly marked as the earlier finite-difference model, distinct from the audited MAC values.
5. **Near-wall note.** A deliberate first-order asymmetry in the v y -diffusion stencil is documented in place (left unchanged to preserve the audited results).

3 Verification results

Table 2 lists independent reruns through the audited driver’s own convergence and root-finding logic. These are reproduction runs (each within the 0.075% root tolerance), reported separately from the final production values. Discrete mass conservation holds to roundoff and the resistance ordering is recovered exactly.

Table 2: Independent rerun verification (corrected code). $Q_0 = 0.0055463$.

Case	P	U_{in}	Q_{rack}	$(Q_{\text{rack}} - Q_0)/Q_0$	Δp	R_h
Small	0.040	0.69372	$5.544\,00 \times 10^{-3}$	-0.041%	0.081714	14.739
Medium	0.060	0.38844	$5.544\,30 \times 10^{-3}$	-0.036%	0.015134	2.7297
Large	0.080	0.23734	$5.545\,80 \times 10^{-3}$	-0.008%	0.005473	0.9868

Table 3: Discrete mass conservation: worst section-flow error $\max_y |Q(y) - Q_{\text{in}}|/Q_{\text{in}}$.

Case	max section-flow error
Small	$1.100\,00 \times 10^{-13} \%$
Medium	$4.400\,00 \times 10^{-14} \%$
Large	$4.400\,00 \times 10^{-14} \%$

Reproduced inlet speeds match the audited production targets (0.69374/0.38871/0.23741) to within the root tolerance, and the resistances reproduce the reported 14.739/2.730/0.987.

4 Flow fields

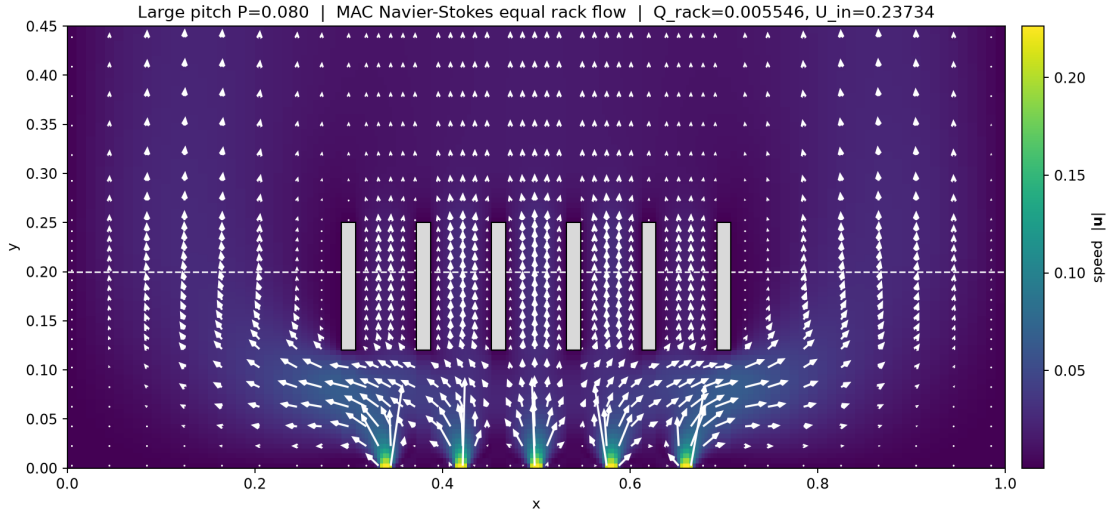


Figure 1: Large-pitch ($P = 0.080$) velocity vector field; colour is speed $|\mathbf{u}|$, white arrows the velocity, gray blocks the wafers, dashed line the $y_{\text{cut}} = 0.20$ flux line.

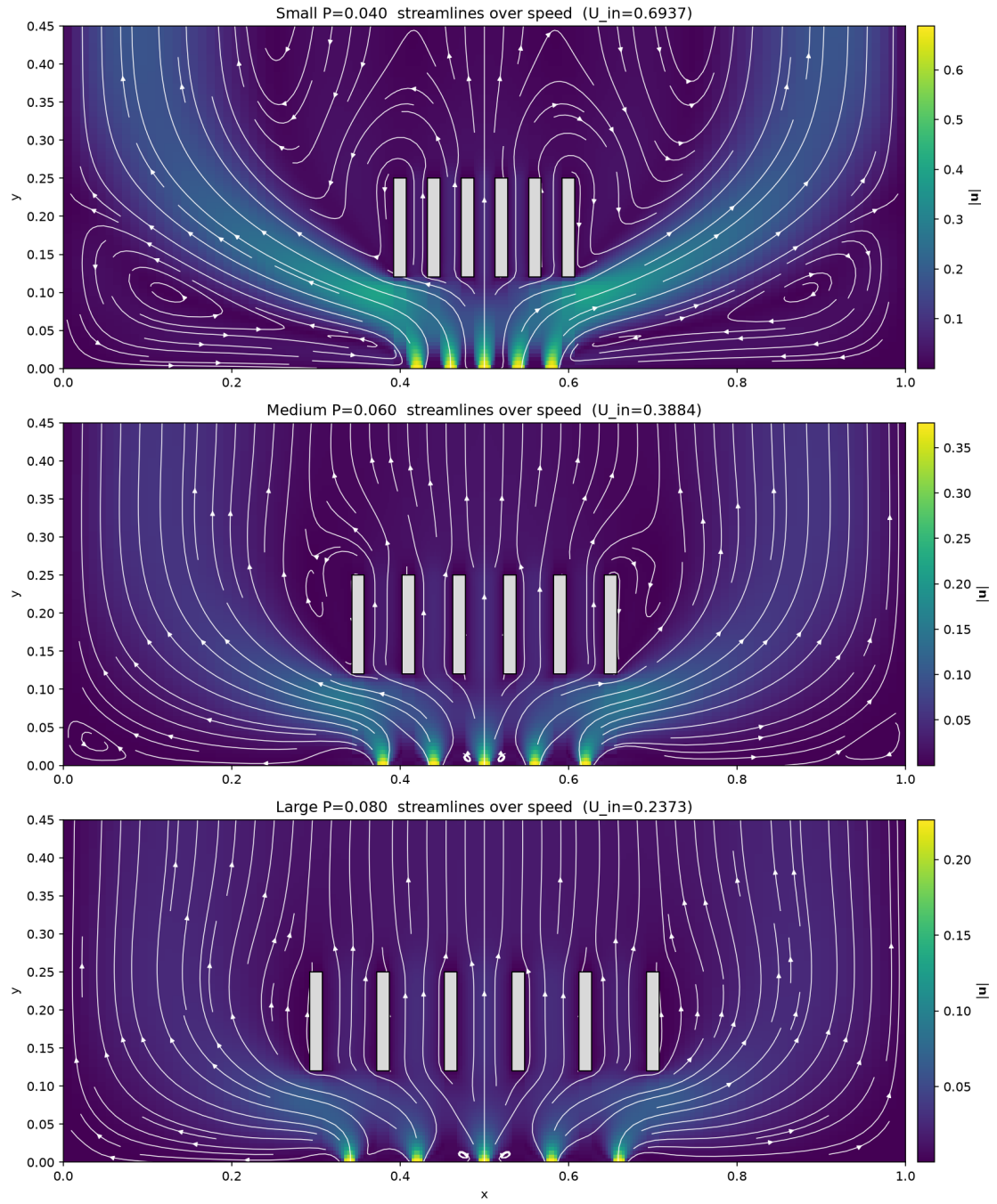


Figure 2: Streamlines over speed for the three pitches (velocity interpolated to a uniform grid for streamline tracing).

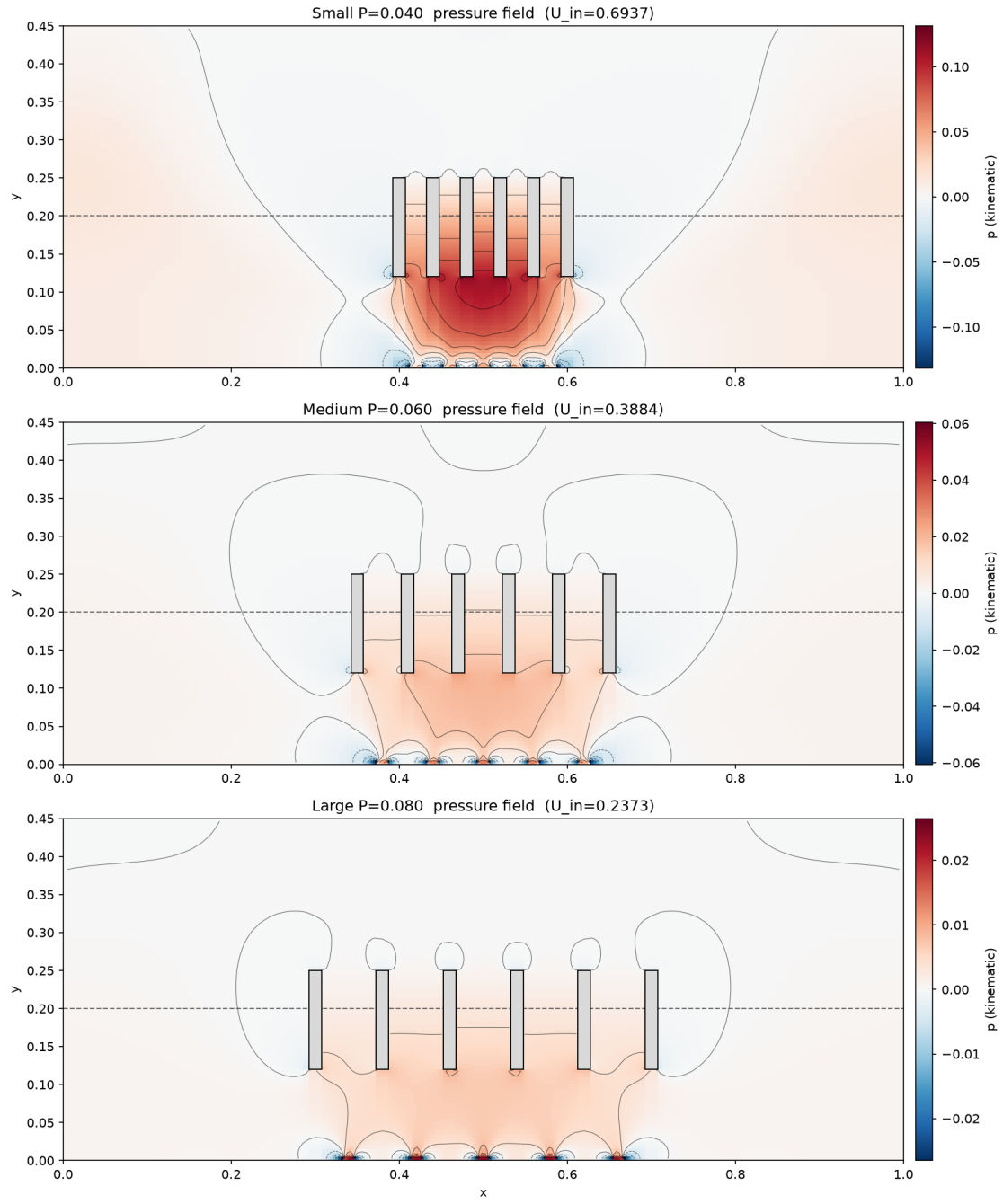


Figure 3: Kinematic pressure field with contours. Pressure drop across the rack grows sharply as the pitch tightens, consistent with the R_h ordering.

5 Equal- Q cut profiles

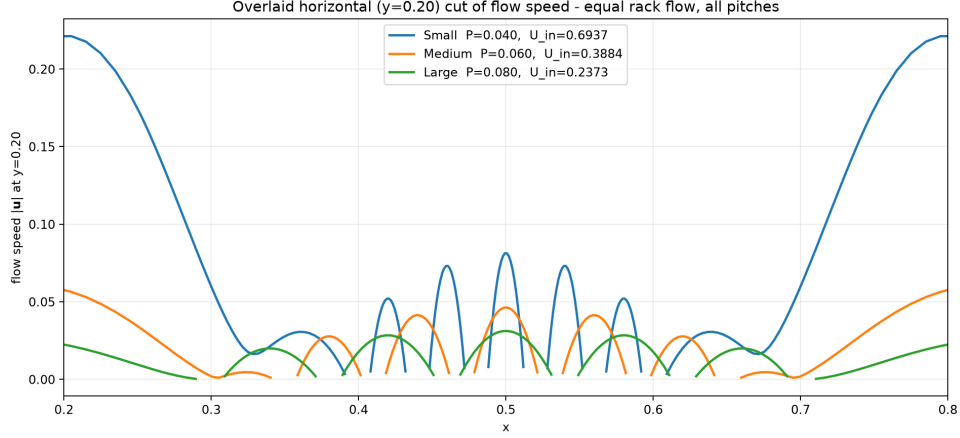


Figure 4: Overlaid $y_{cut} = 0.20$ cut of flow speed. Tighter pitch reaches a higher peak speed in the clear gaps to carry the common Q_{rack} .

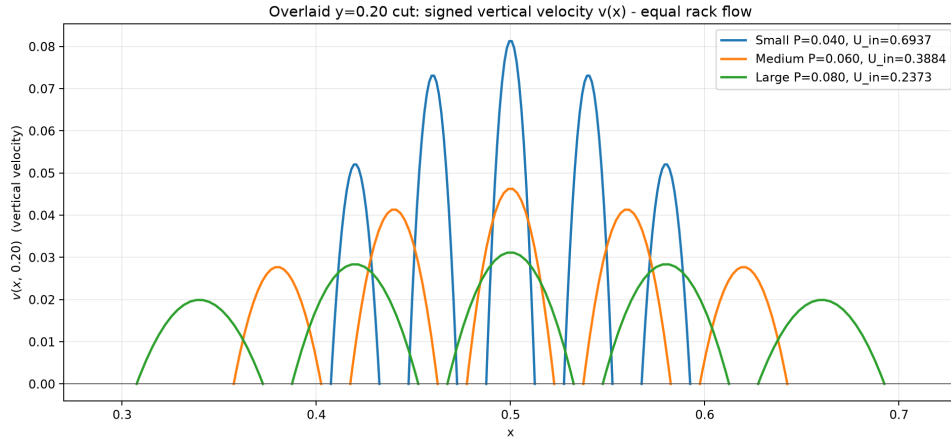


Figure 5: Overlaid $y_{cut} = 0.20$ cut of signed vertical velocity $v(x)$ over the five interior gaps (no-slip zeros imposed at wafer faces).

6 Conclusions

The MAC projection solver is discretely consistent and mass-conserving to roundoff. The equal- Q root solve reliably drives Q_{rack} to the common target, and the corrected, portable code reproduces the audited inlet speeds and hydraulic resistances ($R_h = 14.739 > 2.730 > 0.987$ for Small > Medium > Large). The absolute inlet speeds of the audited MAC model ($\approx 0.694/0.388/0.237$) differ from the legacy finite-difference model's values because the numeric target Q_0 was carried over as a fixed constant rather than being redefined self-consistently for the MAC discretization.